Parametric supervised learning

- **Data:** n observations $(x_i, y_i) \in \mathbb{R}^d \times \mathbb{R}, i = 1, \ldots, n$
- **Prediction function** $h_\theta(x) \in \mathbb{R}$ parameterised by $\theta \in \Theta$

x_1

$y_1 = 1$

x_2

$y_2 = 1$

x_3

$y_3 = -1$

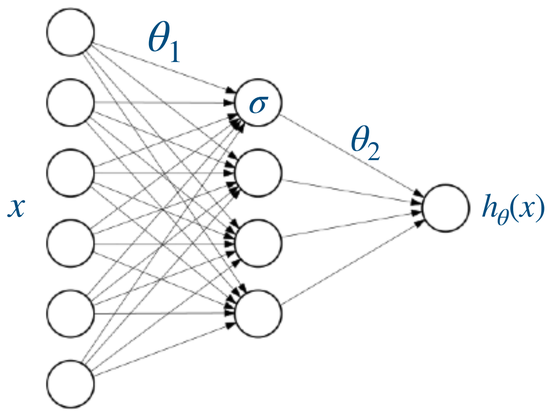
x_4

$y_4 = -1$

Neural networks $\quad h_\theta(x) = \theta_L^\top \sigma\left(\theta_{L-1}^\top \sigma\left(\ldots \theta_2^\top \sigma(\theta_1^\top x)\right)\right)$

Bias term with $\tilde{x} = (x, 1)$

(non-linear) activation



Empirical observations

Cifar-10 dataset:

ResNet-18 (with Batch Norm):

	Train accuracy	**Test accuracy**
GD	100 %	~72% (??)
SGD	100 %	85 %
SGD + momentum	100 %	89 %
SGD + mom + augm + ℓ_2	100 %	95 %
GD + mom + augm + ℓ_2	100 %	87 %

Stochastic Training is Not Necessary for Generalization, Geiping et al. 2021
Bad Global Minima Exist and SGD Can Reach Them, Liu et al. 2020
Revisiting Small Batch Training For Deep Neural Networks, Masters and Luschi 2018